

Labs 8-10: GIS Analyst II

Remote Sensing (7/05)

Spatial Analyst (14/05)

3D Analyst (21/05)

GIS422

Introduction to GIS

Lab Stream 03 (Wednesday)

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1. LAB 8: REMOTE SNEISING

1.1 Summary

1.1.1 Introduction

In this exercise, real geospatial imagery was analysed and processed simulating a real-world project. Multiband/multispectral satellite imagery was used, generating a vegetation index and classifying surface types using clustering algorithms.

1.1.2 Methods

An image of New Zealand was loaded in ArcGIS Pro. Spectral signatures of the Christchurch CBD, Hagley Park, the Lyttelton Harbour, and a snowy peak in the Southern Alps, were each examined. Normalised difference vegetation index (NDVI) was calculated for an area centred on the CBD and the range of values were examined. The remap function was used to classify NDVI values as vegetation or not. The remapped image and histogram are presented as Figures 1-2.

An unsupervised classification tool was used (five classes, default hyperparameters) to generate a raster layer of classes. The majority filter tool and boundary clean tool were applied.

Representations of the layer before and after cleaning are presented in Figure 3. These processes were repeated with a supervised classifier. Areas of vegetation, urban development, and water were identified and labelled as training data. The resulting layers are represented in Figure 4. Each raster was converted to a vector, and the polygons each containing the Christchurch CBD were exported and visualised in Figure 5.

1.1.3 Results

The unsupervised model appeared to capture the CBD much better than the supervised. This was expected, as using only three classes in the supervised model meant that areas that were not urban or forest were misclassified as one of these, and the classified CBD extended too far to the south-west.

1.1.4 Discussion

When applying clustering algorithms in real situations, care should be taken when selecting the number of classes. The supervised algorithm in this exercise had too few classes, and a large amount farmland got classified as urban area, ultimately leading to the overextended classification of the CBD. In future instances, multiple combinations of classes could be tested to see which number produces the desired result, in this case the accurate boundary of the Christchurch CBD.

1.2 Graphics

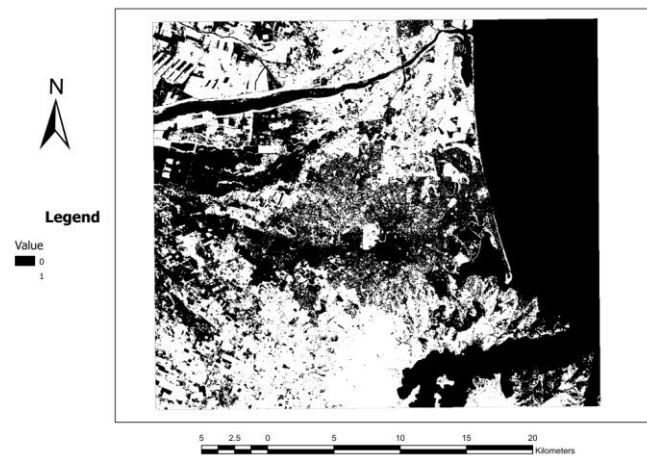


Figure 1: A colour slice image of NVDI values classified as vegetation or not vegetation in the area surrounding the Christchurch CBD.

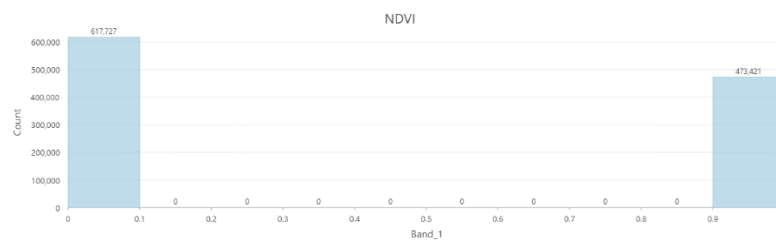


Figure 2: A histogram of the class distribution of the colour slice in Figure 1.

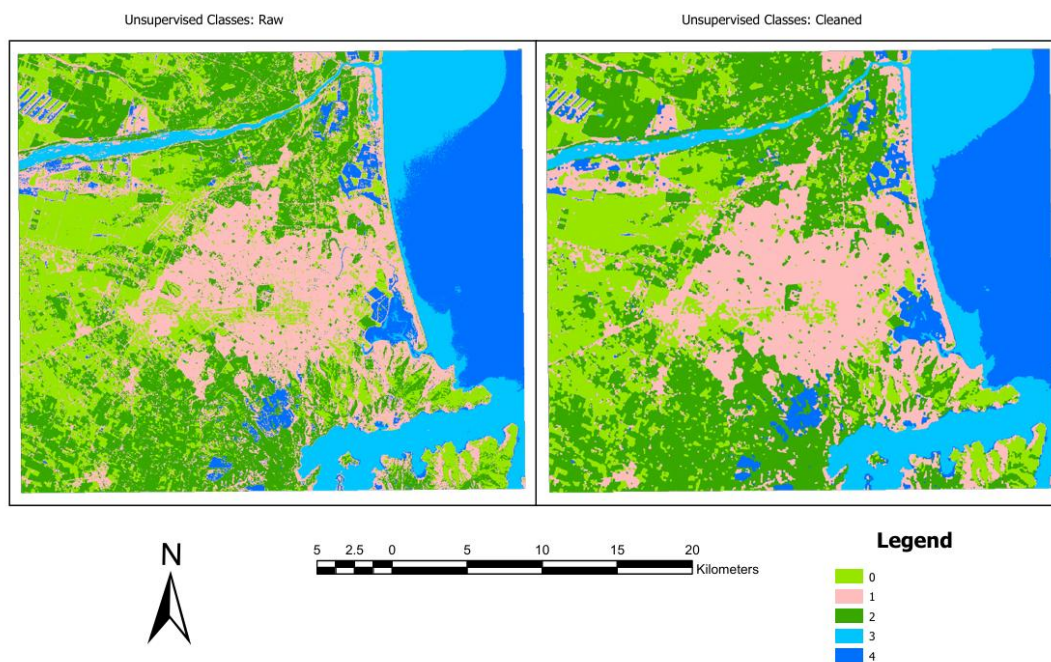


Figure 3: The output of the unsupervised clustering algorithm performed on the area surrounding the Christchurch CBD, before and after cleaning tools were applied.

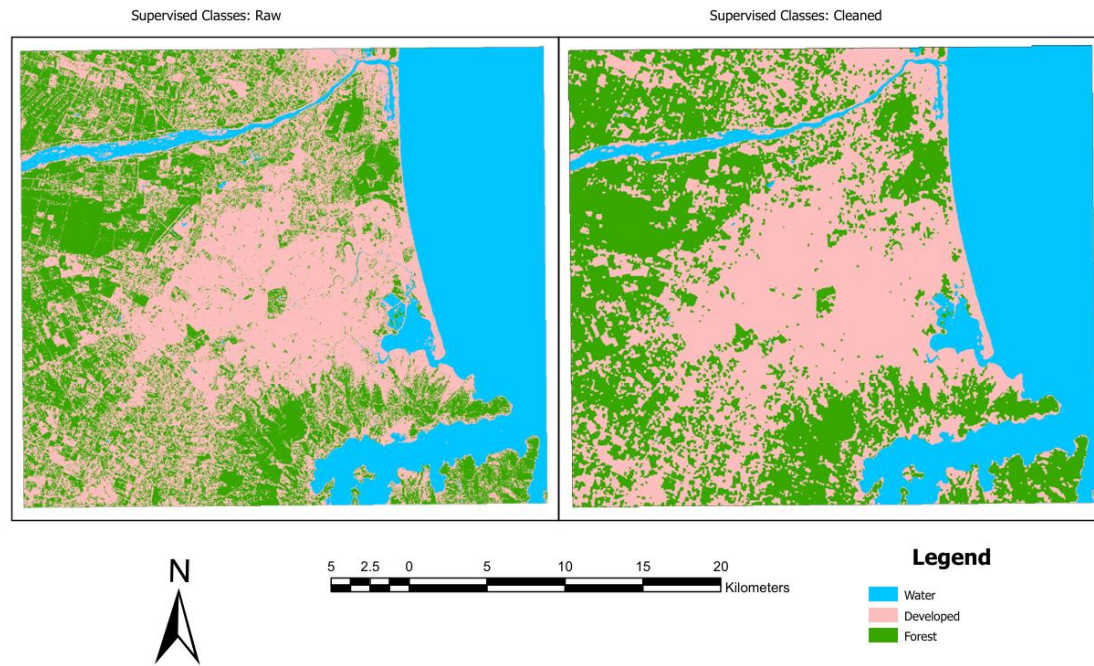


Figure 4: The output of the supervised clustering algorithm performed on the area surrounding the Christchurch CBD, before and after cleaning tools were applied.

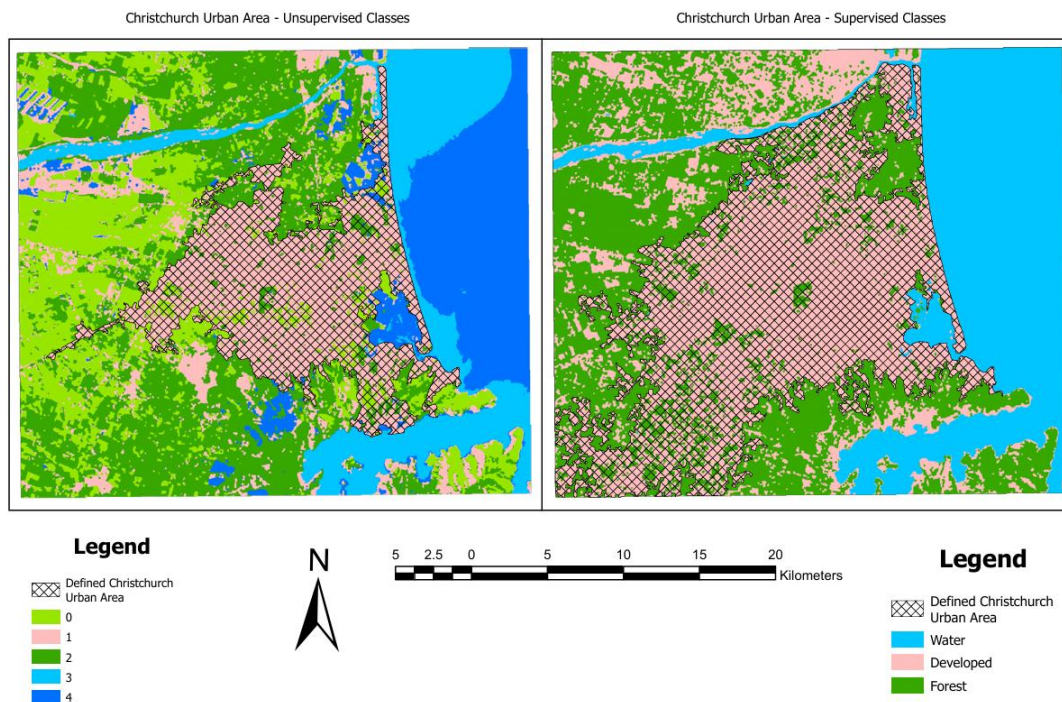


Figure 5: The vector representation of the Christchurch urban area for both clustering models after cleaning.

2. LAB 9: SPATIAL ANALYST

2.1 Summary

2.1.1 Introduction

This laboratory exercise considers the town of Stowe in Vermont, USA. The town has increased significantly in population, and it was decided a new school should be built. The exercise aims to find suitable sites for the school, away from existing schools, close to recreation sites, and on mostly flat ground.

2.1.2 Methods

Two raster layers representing elevation and land use were loaded into ArcGIS Pro alongside locations of schools, recreation sites, and roads. A hillshade was created using the elevation layer, ensuring the Z factor is converted to metric units, and was examined with the land use raster. A bar graph of the classes in the land use layer was produced, such that the class distribution could be examined.

A model was built deriving slope from the elevation dataset and calculating the distance from recreation sites and existing schools to each cell. The three output layers from this model were examined. The model was extended such that the values of these outputs were reclassified to a common measurement scale of 1-10. The weighted overlay tool further extended the model, restricting areas by land type, and weighting the slope and distance measurements, producing a final raster layer with values of suitability. A diagram of the final model is shown in the Graphics section as Figure 6.

2.1.3 Results

Examining the class distribution, it was found that the most dominant land type was forest. Maps of land suitability are presented in Figures 7-8, with the model prioritising distance to recreation sites or schools respectively.

2.1.4 Discussion

The areas that appear to be suitable for a school in both models tend to be close to the recreation sites on the outskirts of the town, in the north-west. While this is a useful tool for assisting in this decision, there are limitations. This model doesn't consider population density, roads, or public transport/accessibility, along with many other factors that should be considered when deciding a location for a new school.

2.2 Graphics

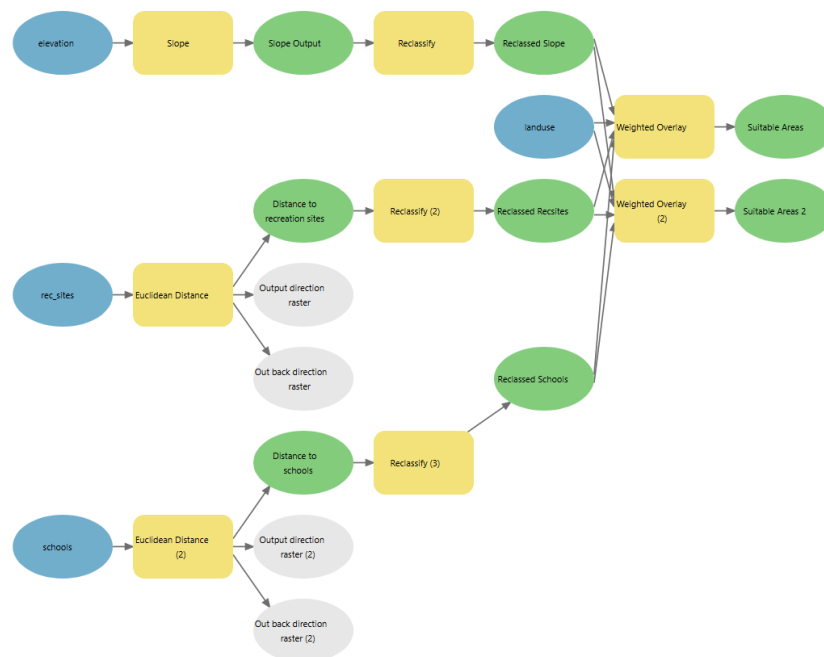
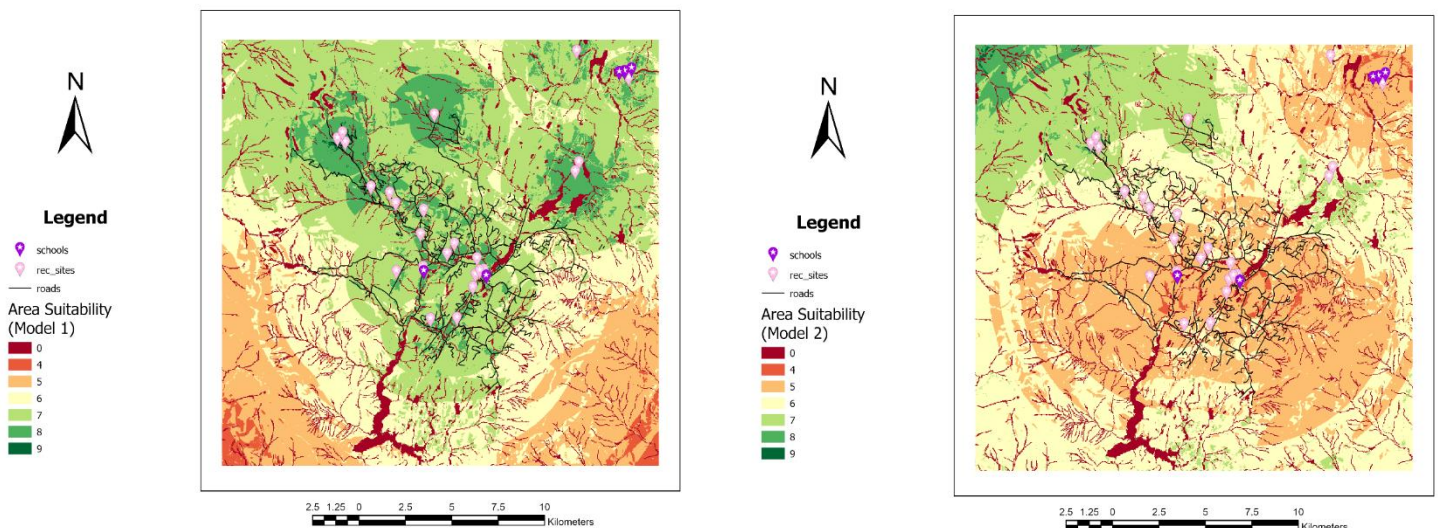


Figure 6: A diagram of the final model used to consider land use, locations of schools, locations of recreation sites, and ground slope to produce a raster representing the suitability of areas for a new school. Instead of running the model twice with different weighting, a second weighted overlay was added with the updated weights.



Figures 7-8: Maps of land suitability for a new school. Model 1 prioritised proximity to recreation sites (left), and model 2 (right) prioritised distance from existing schools.

3. LAB 4: MAP LAYOUTS

3.1 Summary

3.1.1 Introduction

The aim of this laboratory exercise is to investigate the impact of a proposed subdivision in the Port Hills of Christchurch. The existing residents are concerned with losing their views of the ocean. The viewshed of the existing residents will be analysed with and without the subdivision and the impact of the subdivision will be discussed.

3.1.2 Methods

A prebuilt map with features of the area surrounding the proposed subdivision was loaded. A hillshade was constructed from a provided digital elevation model (DEM). The subset of roads representing the concerned residents was exported as a new layer. The viewshed tool was used to produce a raster layer representing the proportion of residents that could observe the land covered by each cell (Figure 9). A new DEM was created, adding the height of the new subdivision building envelope and the viewshed tool was used again. The differences between the output rasters were compared, then the loss of visibility was visualised over the hillshade in a 3D scene, shown in Figures 10-11. An animation of these visualisations was produced and is included with this report as supplementary material.

3.1.3 Results

The output raster layers from the viewshed tool were slightly different, with a very small amount of increased visibility and small sections of decreased visibility surrounding the subdivision.

3.1.4 Discussion

From these visualisations it appears that there will be minimal visual impact to the existing residents due to the proposed subdivision. The reduced area in the viewshed is mostly surrounding the subdivision and only a very small portion is on the shores of the estuary. As the ocean and estuary views were the main concern, it seems that this subdivision should not be held back because of these concerns. The representation of the viewshed is likely to be fairly accurate, and is likely to slightly overrepresent the loss, as buildings will likely not cover the entire building envelope.

3.2 Graphics

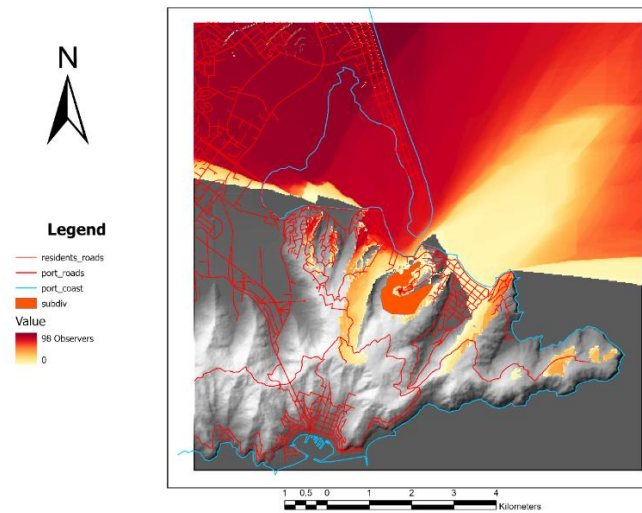


Figure 9: A map layout of the Port Hills area demonstrating the 'view' of the existing residents, before considering the new subdivision, where more red cells represent more observers.

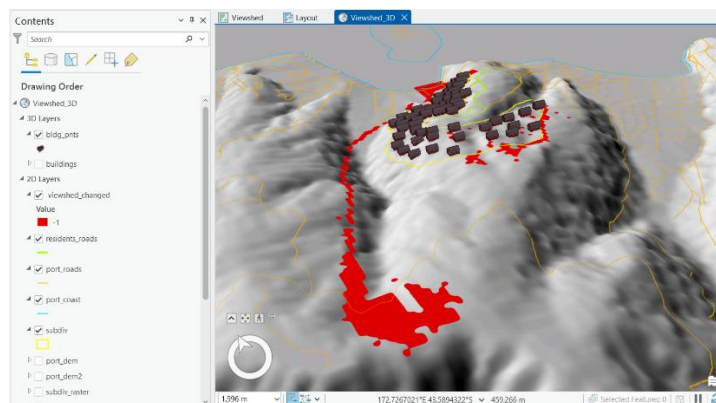


Figure 10: A view looking north of a 'scene' of the proposed subdivision, with the reduced visibility of the existing residents shown in red.

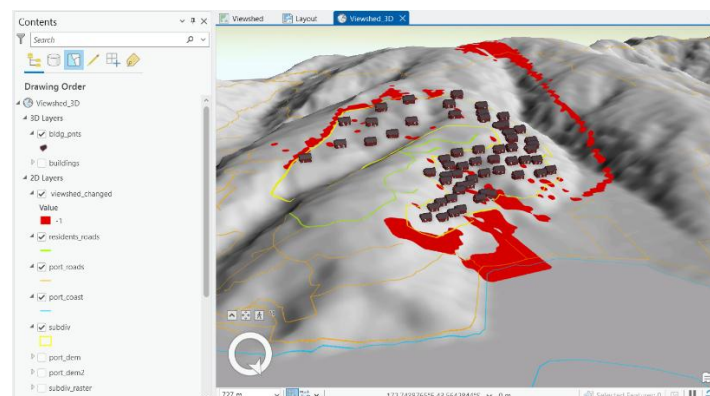


Figure 11: A view looking south of a 'scene' of the proposed subdivision, with the reduced visibility of the existing residents shown in red.